

Jordan University of Science & Technology
Department of Chemistry
CH 101 Second Exam III
30/7/2013

الرقم المتسلسل:

الرقم الجامعي:

الإسم:

الشعبة:

المدرس: د.

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	A	A	D	B	C	B	A	C	A	B	D	D	C	C	A	C	A	A	A	C

1 H 1.008	2 He 4.003																		
3 Li 6.941	4 Be 9.01	5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.18												
11 Na 22.989	12 Mg 24.305	13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948	19 K 39.098	20 Ca 40.08	21 Sc 44.956	22 Ti 47.88	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.847	27 Co 58.933	28 Ni 58.69	29 Cu 63.546	30 Zn 65.39
31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.8	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71
51 Sb 121.75	52 Te 127.6	53 I 126.9	54 Xe 131.29	55 Cs 132.9	56 Ba 137.33	57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm 144.91	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05
71 Lu 174.97	72 Hf 178.49	73 Ta 180.95	74 W 183.85	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po 209	85 At 210	86 Rn 222	87 Fr 223	88 Ra 226.02	89 Ac 227.03	90 Th 232.04

$$h = 6.63 \times 10^{-34} \text{ J-s}$$

Q1) Which of these choices is the electron configuration of an *excited state* of Cr atom?

- A. $[\text{Ar}]4s^23d^5$ B. $[\text{Ar}]4s^13d^5$ C. $[\text{Ar}]4s^23d^6$ D. $[\text{Ar}]4s^23d^4$

Q2) Consider the element with the electron configuration $[\text{Kr}]5s^24d^{10}5p^5$. This element is

- A. representative element. B. transition metal. C. alkali metal. D. actinide element.

Q3) Which of these species make an *isoelectronic pair*: Cl^- , O^{2-} , F^- , Ca^{2+} , Fe^{3+} ?

- A. Ca^{2+} and Fe^{3+} B. O^{2-} and F^- C. F^- and Cl^- D. Cl^- and Ca^{2+}

Q4) Which of these compounds is most likely to be covalent?

- A. KF B. SF_4 C. CaCl_2 D. Al_2O_3

Q5) For which of these reactions is the enthalpy change equal to the second ionization energy of nitrogen?

- A. $\text{N}^+(g) \rightarrow \text{N}^{2+}(g) + e^-$ B. $\text{N}^{2+}(g) + e^- \rightarrow \text{N}^+(g) + e^-$
C. $\text{N}(g) \rightarrow \text{N}^{2+}(g) + 2e^-$ D. $\text{N}^-(g) + e^- \rightarrow \text{N}^{2-}(g)$

Q6) Which of these elements has the greatest metallic character?

- A. Br B. Sc C. Ge D. Mn

Q7) SeO is expected to be a/an _____ oxide.

- A. acidic B. ionic C. amphoteric D. basic

Q8) Which one of the following sets best describes an orbital?

- A. n, l, ms B. n, ml, ms C. n, l, ml D. l, ml, ms

~~Q9~~) Calculate the wavelength of an electron traveling with a velocity of 7.0×10^3 kilometers per second. The mass of an electron is 9.1×10^{-28} g.

- A. 1.0×10^{-10} m B. 1.0×10^{-7} m C. 1.0 m D. 1.0×10^{-13} m

Q10) What is the maximum number of electrons in an atom that can have the following set of quantum numbers? $n = 3$ $l = 2$ $m_l = -2$ $m_s = +1/2$

- A. 5 B. 1 C. 2 D. 10

Q11) Which one of the following molecules has tetrahedral geometry?

- A. XeF₄ B. BF₃ C. AsF₅ D. CF₄

Q12) Complete this sentence: The PCl₅ molecule has

- A. nonpolar bonds, and is a nonpolar molecule. B. nonpolar bonds, but is a polar molecule.
C. polar bonds, and is a polar molecule. D. polar bonds, but is a nonpolar molecule.

Q13) The cobalt(III) ion, Co³⁺, has how many 3d electrons?

- A. 0 B. 7 C. 6 D. 5

~~Q14~~) How many orbitals are allowed in a subshell if the angular momentum quantum number for electrons in that subshell is 3?

- A. 1 B. 3 C. 5 D. 7

Q15) "No two electrons in an atom can have the same four quantum numbers" is a statement of

- A. Pauli exclusion principle B. Bohr's equation. C. Hund's rule. D. Aufbau principle

Q16) Transition metal elements have atoms or ions with partially filled

- A. s subshells. B. p subshells. C. d subshells. D. f subshells.

~~Q17~~) The Lewis dot symbol for the S²⁻ ion is

- A. $\cdot\ddot{S}\cdot$ B. $\cdot\ddot{S}:^{2-}$ C. S^{2-} D. $\text{---}\ddot{S}\text{---}^{2-}$

Q18) Which of these elements is the least electronegative?

- A. Sr B. V C. Ni D. P

Q19) The energy change for the reaction $A(g) + B(g) \rightarrow A^-(g) + B^+(g)$ is ?

- A. IE(A) - EA(B) B. IE(A) + EA(B) C. EA(B) - IE(A) D. -(EA(B) + IE(A))

Q20) Which of these compounds does not follow the octet rule?

- A. NF₃ B. CF₄ C. PF₅ D. AsH₃